

ERNIE 3D printing instructions.

Requirements

Before printing the ERNIE components, ensure that the following requirements are met:

- A **PLA-compatible 3D printer** is required. The ERNIE components were printed using a [Bambu Lab 3D printer](#). If using the same printer, it can be purchased from the manufacturer's website.
- Install [Bambu Studio](#) software, which is used to open, prepare, and generate print files from the provided .stl models.
- PLA filament is recommended for printing the ERNIE components.

3D Printing Procedure

Step 1: Install Bambu Studio Software

Download and install [Bambu Studio](#) before accessing the ERNIE 3D model files. The software is required to open, prepare, slice, and generate printing files from the provided .stl models.

Step 2: Download ERNIE 3D Model Files

Download the required [ERNIE component files](#) from the GitHub repository.

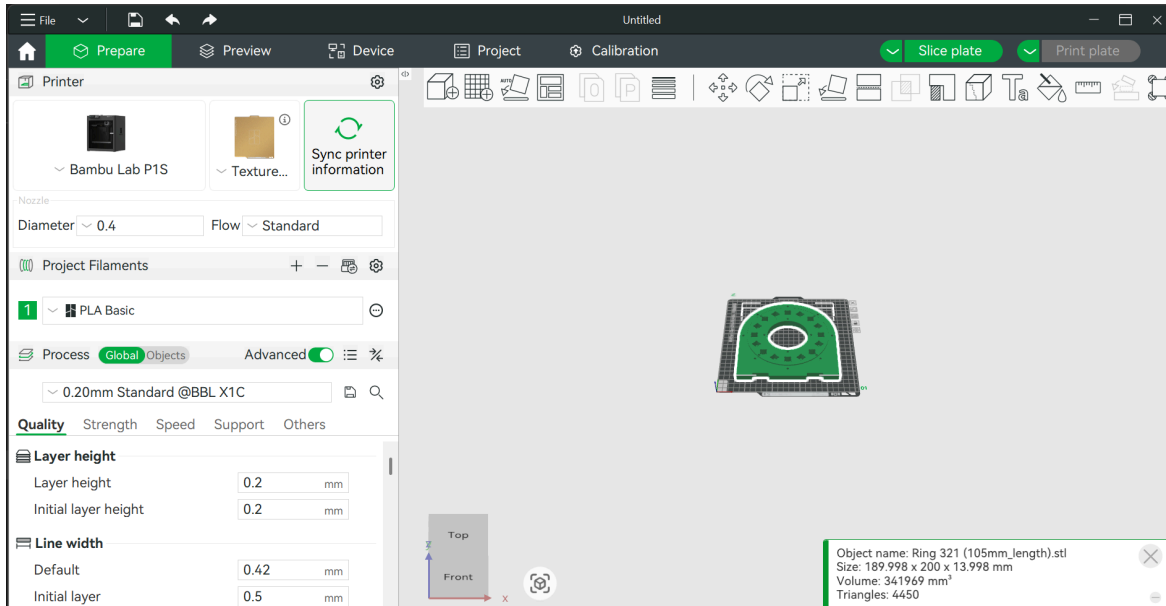
Step 3: Prepare the SD Card

Remove the SD card from the 3D printer and insert it into the computer. The generated printing files will be transferred to this SD card before loading them onto the printer.

Step 4: Open the STL File

Locate the required [.stl](#) file and open it using Bambu Studio.

The model should appear as shown below:



Step 5: Configure Printer and Material Settings

In the **Global Settings** panel on the left side of the Bambu Studio interface:

- Select the appropriate printer model from the printer drop-down menu.
- Select **Bambu PLA Basic** as the filament material.
- Select the printing process preset. The **0.20 mm Standard** profile is recommended for general PLA printing.

Step 6: Configure Quality and Strength Settings

Navigate to the **Quality** tab and verify the layer height.

Under the **Strength** tab, use the following recommended settings:

- **Infill density:** 15–20%
- **Infill pattern:** Grid
- **Supports:** Not required for most ERNIE ring components.

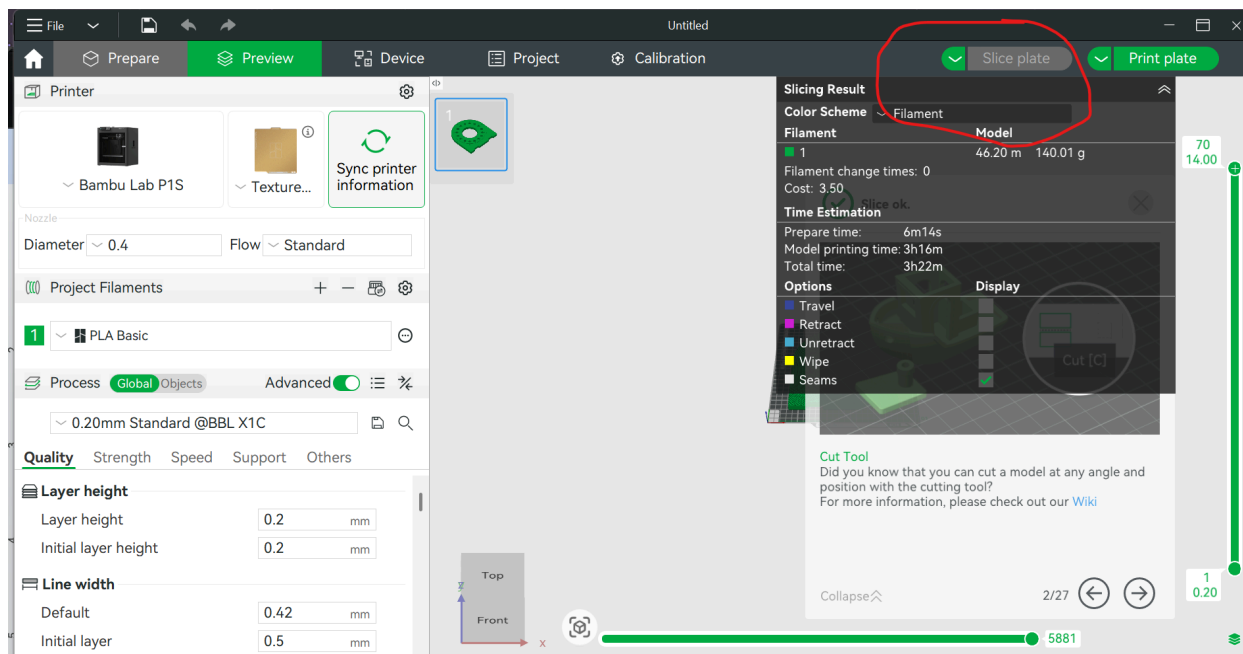
Step 7: Slice and Preview the Model

Select **Slice Plate** in the upper-right corner of Bambu Studio.

Allow the software to generate the printing paths and layer information. Use the preview slider to inspect the model layer-by-layer.

Verify:

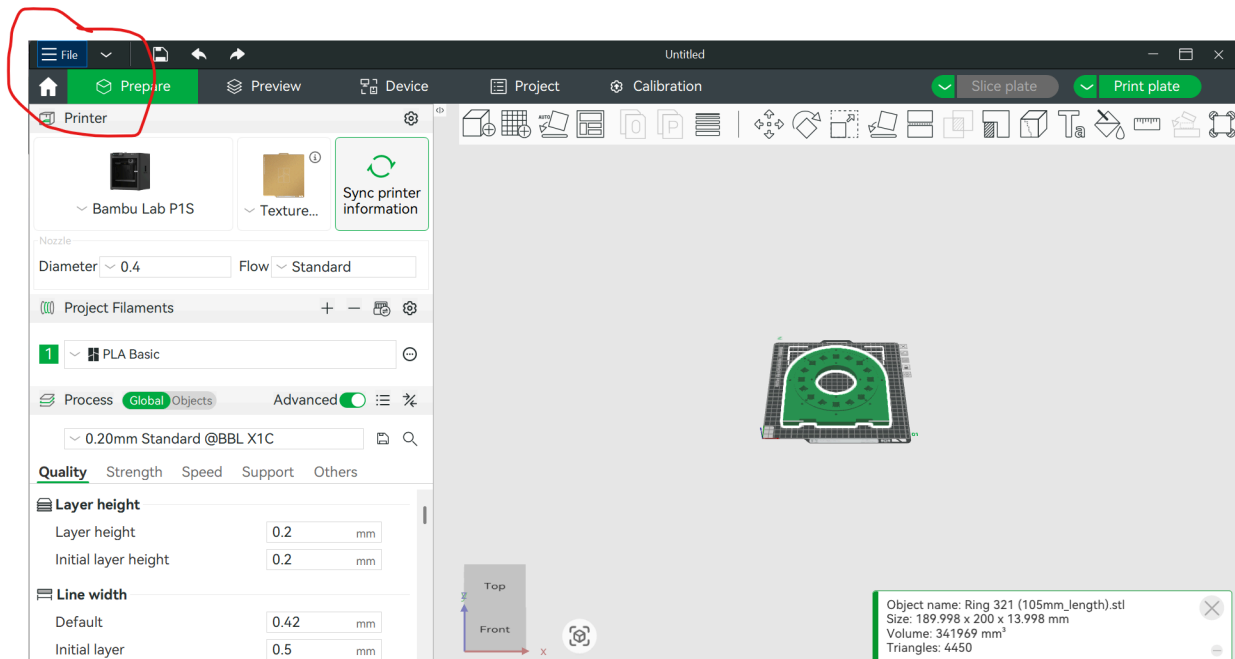
- Estimated printing time
- Filament usage
- Correct model preparation



Step 8: Export the Printing File

From the top-left menu: **File** → **Export** → **Export G-code**

Save the generated file onto the SD card.



Step 9: Start Printing

- Load the printer with PLA filament.
- Turn on the 3D printer.
- Insert the SD card containing the G-code file.
- Select the desired ERNIE component.
- Confirm the selection to begin printing.

Step 10: Remove Printed Components

After printing is complete:

1. Allow the printer and component to cool for approximately **5 minutes**.
2. Carefully remove the printed component from the build plate.
3. Repeat the process for all remaining ERNIE components.